

PHYS 350: Honours Electricity and Magnetism

W.R.YCH.

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Abstract

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1 Introduction

Textbook used is David J. Griffiths - Introduction to Electrodynamics Fifth Edition. Course grading is divided as follows:

- 35% problem sets (must be handwritten scans).
- 25% midterm.
- 40% final exam.

2 Vector Analysis

2.1 Terminologies

Definition 1 (Scalar). *A scalar is a quantity with just a magnitude.*

Definition 2 (Vector). *A vector is a quantity with a magnitude and a direction, i.e. $\vec{v} = v_x\hat{x} + v_y\hat{y} + v_z\hat{z}$.*

Definition 3 (Cross Product). *The cross product is a vector perpendicular to both, set by the right hand rule, and it vanishes for parallel vectors:*

$$\vec{v} \times \vec{w} = |\vec{v}||\vec{w}| \sin \theta \hat{n} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix} = \hat{x}(v_y w_z - w_y v_z) - \hat{y}(v_x w_z - w_x v_z) + \hat{z}(v_x w_y - w_x v_y). \quad (1)$$

Definition 4 (Dot Product). *The dot product measures how much of $\vec{v}$ points along $\vec{w}$, so it is zero for perpendicular vectors:*

$$\vec{v} \cdot \vec{w} = v_x w_x + v_y w_y + v_z w_z = |\vec{v}||\vec{w}| \cos \theta. \quad (2)$$

Remark 1. Note $\vec{v} \times \vec{w} = -\vec{w} \times \vec{v}$, and $|\vec{v}| = \sqrt{\vec{v} \cdot \vec{v}}$.

Definition 5 (Field). *A field is something with a value at every point in space. A scalar field assigns a number to each point, like temperature $T(x, y, z)$ or the height of a hill. A vector field assigns a vector to each point, like fluid velocity or $\vec{E}$.*

2.2 Differential Calculus

Definition 6 (Del operator). *The del operator*

$$\vec{\nabla} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right).$$

Definition 7 (Gradient). *The gradient turns a scalar into a vector, pointing in the direction of steepest increase of f :*

$$\vec{\nabla} f = \frac{\partial f}{\partial x} \hat{x} + \frac{\partial f}{\partial y} \hat{y} + \frac{\partial f}{\partial z} \hat{z}. \quad (3)$$

Remark 2. The gradient points in the steepest uphill direction, and its length grows where the hill is steeper. It is always perpendicular to the contours of constant f .

Definition 8 (Divergence). Divergence turns a vector into a scalar, measuring sources and sinks. For a small box, non-zero divergence means the inflow and outflow through its faces do not balance: arrows spreading from a point (positive) mean fluid is created there, arrows converging (negative) mean it is destroyed.

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}. \quad (4)$$

Definition 9 (Curl). Curl turns a vector into a vector, measuring local rotation. Picture a small paddle wheel: if the flow spins it, the curl is non-zero there. Uniform flow does not spin it, but shear flow (parallel arrows, longer on one side) does, so straight field lines do not imply zero curl.

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} = \left(\frac{\partial}{\partial y} F_z - \frac{\partial}{\partial z} F_y, -\left(\frac{\partial}{\partial x} F_z - \frac{\partial}{\partial z} F_x \right), \frac{\partial}{\partial x} F_y - \frac{\partial}{\partial y} F_x \right). \quad (5)$$

Remark 3. The direction of the curl follows the right-hand rule: curl your fingers along the paddle wheel's rotation and your thumb points along $\vec{\nabla} \times \vec{F}$.

Remark 4. Divergence and curl are themselves fields: they can vanish in some places and not others, and a general field has both.

2.3 Integral Calculus

Definition 10 (Integral). An integral is an instruction to move along a region, evaluate something at each step, and add it to a running total.

Definition 11 (Line Integral). A line integral sums $\vec{F}$ along a curve, picking up only the component tangent to the path:

$$\int_C \vec{F} \cdot d\vec{l}.$$

Example. Take $\vec{F} = y\hat{x}$ along a counterclockwise quarter circle of radius R .

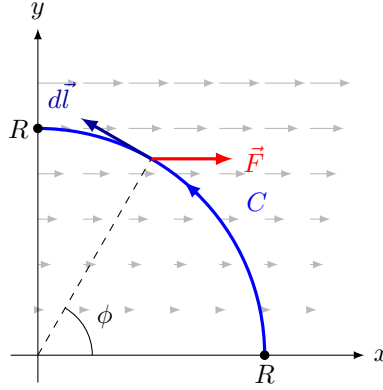


Figure 1: The field $\vec{F} = y\hat{x}$ (gray) and the quarter circle C . At a typical point, $\vec{F}$ points along $+\hat{x}$ while $d\vec{l}$ has a $-\hat{x}$ component, so $\vec{F} \cdot d\vec{l} < 0$.

On the circle $d\vec{l} = \frac{d\vec{r}}{d\phi} d\phi = R(-\sin\phi\hat{x} + \cos\phi\hat{y})d\phi$ and $y = R\sin\phi$, so

$$\int_C \vec{F} \cdot d\vec{l} = \int_0^{\pi/2} \begin{pmatrix} R\sin\phi \\ 0 \end{pmatrix} \cdot R \begin{pmatrix} -\sin\phi \\ \cos\phi \end{pmatrix} d\phi = -R^2 \int_0^{\pi/2} \sin^2\phi d\phi = -\frac{R^2\pi}{4}.$$

It is negative because $\vec{F}$ points along $+\hat{x}$ while we travel in the $-\hat{x}$ direction over most of the arc.

Definition 12 (Flux). Flux measures how much of the field pokes through a surface:

$$\int_S \vec{F} \cdot d\vec{a} = \int_S \vec{F} \cdot \hat{n} da.$$

Remark 5. Flux is the total amount of a vector field flowing through a surface. Divergence is the flux per unit volume at a specific point.

Definition 13 (Volume Integral). A volume integral sums a scalar over a region¹:

$$\int_V f d^3\vec{r} = \int_V f dx dy dz.$$

Remark 6. We write $\oint$ when the curve is a closed loop or the surface is closed.

2.3.1 The Three Fundamental Theorems

Integrating a derivative over a region reduces to evaluating the original object on the boundary of that region.

¹A volume integral gives the mass if f is a density

Theorem 1 (Gradient theorem). *The boundary of a curve is two points:*

$$\int_C \vec{\nabla} h \cdot d\vec{l} = h(b) - h(a).$$

To find my altitude change climbing a hill, I can compare before and after, or add up all the little changes along the way.

Theorem 2 (Stokes' theorem). *The boundary of a surface is a curve:*

$$\iint_S (\vec{\nabla} \times \vec{F}) \cdot d\vec{S} = \oint_C \vec{F} \cdot d\vec{l}. \quad (6)$$

To get the total swirl, I can work out the rotation at the boundary, or add up all the little whirlpools inside. Neighbouring whirlpools cancel along their shared edges, leaving only the boundary.

Theorem 3 (Gauss' divergence theorem). *The boundary of a volume is a surface:*

$$\oiint_S (\vec{F} \cdot \hat{n}) dS = \iiint_V (\vec{\nabla} \cdot \vec{F}) dV. \quad (7)$$

To find the total flux leaking out through the surface, I can add up all the sources inside. Flux between neighbouring cells cancels, leaving only the outer surface.

2.4 Index Notation

Remark 7. *A matrix carries two indices, A_{ij} , where i says which row and j says which column.*

Definition 14 (Transpose). *The transpose swaps the indices, $(A^T)_{ij} = A_{ji}$.*

Definition 15 (Symmetry). *A symmetric matrix has $A_{ij} = A_{ji}$, and an anti-symmetric one has $B_{ij} = -B_{ji}$.*

Definition 16 (Kronecker delta). *The Kronecker delta equals +1 when its condition is fulfilled and 0 otherwise, so it could be the identity matrix in index notation:*

$$\delta_{ij} = \begin{cases} +1 & i = j \\ 0 & i \neq j \end{cases}.$$

Example.

$$\sum_j \delta_{ij} v_j = v_i.$$

Proposition 1 (Matrix Multiplication).

$$\vec{w} = A\vec{v} \implies w_i = \sum_j A_{ij} v_j.$$

Proposition 2 (Dot Product).

$$\vec{v} \cdot \vec{w} = \sum_i v_i w_i; \vec{\nabla} \cdot \vec{F} = \sum_i \partial_i F_i.$$

Proposition 3 (Cross Product).

$$(\vec{v} \times \vec{w})_i = \sum_{j,k} \varepsilon_{ijk} v_j w_k; (\vec{\nabla} \times \vec{F})_i = \sum_{j,k} \varepsilon_{ijk} \partial_j F_k.$$

Definition 17 (Levi-Civita symbol). ε_{ijk} is defined by

- $\varepsilon_{123} = \varepsilon_{231} = \varepsilon_{312} = +1$.
- $\varepsilon_{321} = \varepsilon_{213} = \varepsilon_{132} = -1$.
- If any two indices are equal, $\varepsilon_{ijk} = 0$.

Example. For $i = 1$, only the terms with $(j, k) = (2, 3)$ and $(3, 2)$ survive:

$$(\vec{v} \times \vec{w})_1 = \varepsilon_{123} v_2 w_3 + \varepsilon_{132} v_3 w_2 = v_2 w_3 - v_3 w_2.$$

Proposition 4 (Epsilon-delta identity). Contracting two Levi-Civita symbols on a shared index gives

$$\sum_i \varepsilon_{ijk} \varepsilon_{imn} = \delta_{jm} \delta_{kn} - \delta_{jn} \delta_{km}.$$

2.5 Second Derivatives Identities

Definition 18 (Laplacian). Taking the divergence of a gradient gives

$$\vec{\nabla} \cdot (\vec{\nabla} f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \equiv \nabla^2 f, \quad (8)$$

where ∇^2 is the Laplacian².

Remark 8. ∇^2 also acts on vector fields, component by component: $(\nabla^2 \vec{F})_i = \nabla^2 F_i$.

The full list of second derivative identities:

Name	Identity
Divergence of a gradient	$\vec{\nabla} \cdot (\vec{\nabla} f) = \nabla^2 f$
Curl of a gradient	$\vec{\nabla} \times (\vec{\nabla} f) = 0$
Divergence of a curl	$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{F}) = 0$
Curl of a curl	$\vec{\nabla} \times (\vec{\nabla} \times \vec{F}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{F}) - \nabla^2 \vec{F}$

Table 1: Second derivative identities.

²Mathematicians often write Δ , but in physics we use ∇^2 .

2.5.1 Proving the Identities

Curl of a Curl. Set $\vec{G} \equiv \vec{\nabla} \times (\vec{\nabla} \times \vec{F})$. Then,

$$G_i = \sum_{j,k} \varepsilon_{ijk} \partial_j \sum_{m,n} \varepsilon_{kmn} \partial_m F_n = \sum_{j,k,m,n} \varepsilon_{ijk} \varepsilon_{kmn} \partial_j \partial_m F_n.$$

Using the cyclic permutation $\varepsilon_{ijk} = \varepsilon_{kij}$, so that the summed index k sits first in both epsilons, the epsilon-delta identity [4](#) gives

$$\sum_k \varepsilon_{kij} \varepsilon_{kmn} = \delta_{im} \delta_{jn} - \delta_{in} \delta_{jm}.$$

Reading off the proposition with the summed index k removed:

$$G_i = \sum_{j,m,n} (\delta_{im} \delta_{jn} - \delta_{in} \delta_{jm}) \partial_j \partial_m F_n.$$

Each Kronecker delta collapses one sum by setting its indices equal. In the first term δ_{im} forces $m = i$ and δ_{jn} forces $n = j$; in the second, δ_{in} forces $n = i$ and δ_{jm} forces $m = j$:

$$G_i = \sum_j \partial_j \partial_i F_j - \sum_j \partial_j \partial_j F_i \stackrel{3}{=} \partial_i \sum_j \partial_j F_j - \sum_j \partial_j \partial_j F_i = \left(\vec{\nabla} (\vec{\nabla} \cdot \vec{F}) \right)_i - (\nabla^2 \vec{F})_i.$$

$$\therefore \vec{\nabla} \times (\vec{\nabla} \times \vec{F}) = \vec{\nabla} (\vec{\nabla} \cdot \vec{F}) - \nabla^2 \vec{F}.$$

□

Curl of a Gradient. Let f be twice continuously differentiable. With $(\vec{\nabla} f)_k = \partial_k f$,

$$\left(\vec{\nabla} \times (\vec{\nabla} f) \right)_i = \sum_{j,k} \varepsilon_{ijk} \partial_j \partial_k f.$$

The indices j and k are summed over, so they can be renamed $j \leftrightarrow k$:

$$\left(\vec{\nabla} \times (\vec{\nabla} f) \right)_i = \sum_{j,k} \varepsilon_{ikj} \partial_k \partial_j f.$$

Now use antisymmetry, $\varepsilon_{ikj} = -\varepsilon_{ijk}$, and the fact that partial derivatives commute, $\partial_k \partial_j f = \partial_j \partial_k f$:

$$\left(\vec{\nabla} \times (\vec{\nabla} f) \right)_i = - \sum_{j,k} \varepsilon_{ijk} \partial_j \partial_k f = - \left(\vec{\nabla} \times (\vec{\nabla} f) \right)_i.$$

A number equal to its own negative is zero, so every component vanishes and $\vec{\nabla} \times (\vec{\nabla} f) = \vec{0}$. □

³This equality uses $\partial_j \partial_i = \partial_i \partial_j$, valid for $\vec{F}$ twice continuously differentiable.

2.6 The Theory of Vector Fields

Theorem 4 (Helmholtz). *If $\vec{\nabla} \cdot \vec{F}$ and $\vec{\nabla} \times \vec{F}$ are specified everywhere and $\vec{F} \rightarrow 0$ at infinity, then $\vec{F}$ is uniquely determined. It can be written as $\vec{F} = \vec{\nabla}\psi + \vec{\nabla} \times \vec{A}$, an irrotational part plus a solenoidal part.*

Theorem 5 (Irrotational fields). *Let $\vec{F}$ be a C^1 vector field on a simply connected domain (e.g. all of $\mathbb{R}^3$). The following are equivalent:*

- (i) $\vec{\nabla} \times \vec{F} = 0$ everywhere.
- (ii) $\int_a^b \vec{F} \cdot d\vec{l}$ is independent of path, for any given end points.
- (iii) $\oint_C \vec{F} \cdot d\vec{l} = 0$ for every closed loop C .
- (iv) $\vec{F} = -\vec{\nabla}\psi$ for some scalar field ψ .

Theorem 6 (Solenoidal fields). *Let $\vec{F}$ be a C^1 vector field on a domain with no enclosed voids (e.g. all of $\mathbb{R}^3$). The following are equivalent:*

- (i) $\vec{\nabla} \cdot \vec{F} = 0$ everywhere.
- (ii) $\int_S \vec{F} \cdot d\vec{a}$ is independent of surface, for any given boundary line.
- (iii) $\oint_S \vec{F} \cdot d\vec{a} = 0$ for any closed surface S .
- (iv) $\vec{F} = \vec{\nabla} \times \vec{A}$ for some vector field $\vec{A}$.

Remark 9. *In electrostatics we will find $\vec{\nabla} \times \vec{E} = 0$, which licenses $\vec{E} = -\vec{\nabla}V$ and the electric potential. In magnetostatics we will find $\vec{\nabla} \cdot \vec{B} = 0$, which licenses $\vec{B} = \vec{\nabla} \times \vec{A}$, the vector potential.*

3 Electrostatics

3.1 Electric Field

Law 1 (Coulomb's law). *The force on a point charge q_2 due to a point charge q_1 is*

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}, \quad (9)$$

where $\hat{r}$ points from q_1 to q_2 .⁴

Definition 19 (Permittivity of free space). *The permittivity of free space measures how easily an electric field can form in a classical vacuum:*

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N} \cdot \text{m}^2).$$

⁴Coulomb's law has the same shape as Newton's gravity. The reason is geometric rather than a coincidence, and Gauss' law makes that explicit: in d spatial dimensions the flux spreads over a $(d-1)$ -dimensional sphere, so $F \propto 1/r^{d-1}$, and the 4π is the area of the unit sphere in $d=3$.

Definition 20 (Electric field). *The electric field is the force per unit test charge. For a set of point charges q_i ,*

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i^2} \hat{\mathbf{r}}_i, \quad \vec{\mathbf{r}}_i \equiv \vec{r} - \vec{r}_i. \quad (10)$$

Principle 1 (Superposition). *The field of several sources is the sum of their individual fields, $\vec{E} = \vec{E}_1 + \vec{E}_2 + \dots$*

Remark 10 (Field theory). *Source charges create $\vec{E}$; a test charge Q then feels $\vec{F} = Q\vec{E}$.*

3.1.1 Continuous Charge Distributions

Definition 21 (Continuous distributions). *Replacing q_i by $dq = \lambda dl' = \sigma da' = \rho d\tau'$ turns the sum into an integral:*

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r^2} \hat{\mathbf{r}}, \quad \vec{\mathbf{r}} = \vec{r} - \vec{r}', \quad (11)$$

where $\vec{r}$ is the field point and $\vec{r}'$ the source point.

Remark 11. $\hat{\mathbf{r}}$ is not constant: it points differently for every source point $\vec{r}'$, so it cannot come out of the integral. Split it into components first.

Example (Line of charge). *A straight segment of length $2L$, centred at the origin on the x axis, carries a uniform line charge density λ . Find $\vec{E}$ at the point $P = (0, 0, z)$, a height z above its midpoint.*

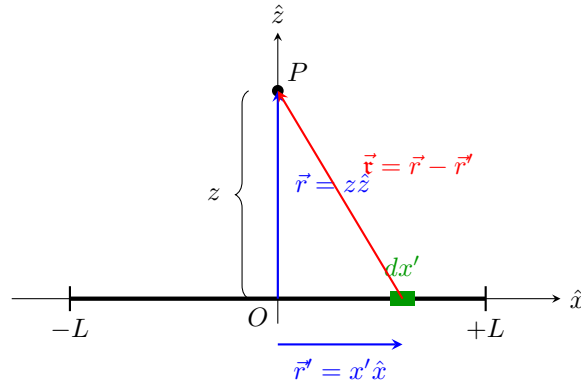


Figure 2: Setup for the finite line of charge. The source point $\vec{r}' = x'\hat{x}$ runs from $-L$ to L ; the field point is P at $\vec{r} = z\hat{z}$; the separation vector is $\vec{\mathbf{r}} = \vec{r} - \vec{r}'$.

With $\vec{r} = z\hat{z}$ and $\vec{r}' = x'\hat{x}$,

$$\vec{\mathbf{r}} = z\hat{z} - x'\hat{x}, \quad r = \sqrt{z^2 + x'^2},$$

so

$$\vec{E} = \frac{\lambda}{4\pi\epsilon_0} \int_{-L}^L \frac{z\hat{z} - x'\hat{x}}{(z^2 + x'^2)^{3/2}} dx' = \frac{\lambda}{4\pi\epsilon_0} \left[\hat{z} \int_{-L}^L \frac{z}{(z^2 + x'^2)^{3/2}} dx' - \hat{x} \int_{-L}^L \frac{x'}{(z^2 + x'^2)^{3/2}} dx' \right].$$

The second integral vanishes by symmetry: its integrand is odd in x' , so the pieces at x' and $-x'$ cancel. The first is done with the substitution $x' = z \tan \theta$, giving

$$\vec{E} = \frac{\lambda}{4\pi\epsilon_0} \frac{1}{z} \left[\frac{x'}{\sqrt{x'^2 + z^2}} \right]_{-L}^L \hat{z} = \frac{\lambda}{4\pi\epsilon_0} \frac{2L}{z\sqrt{L^2 + z^2}} \hat{z} = \frac{\lambda}{2\pi\epsilon_0 z} \cdot \frac{L}{\sqrt{L^2 + z^2}} \hat{z}.$$

This has two limits:

- Close to the line ($z \ll L$), it looks like an infinite line: $\vec{E} \rightarrow \frac{\lambda}{2\pi\epsilon_0 z} \hat{z}$.
- Far from the line ($z \gg L$), it looks like a point charge: $\vec{E} \rightarrow \frac{Q}{4\pi\epsilon_0 z^2} \hat{z}$ with $Q = 2\lambda L$.

Remark 12. Sanity check: an answer for an extended distribution should reduce to a simpler known field in the limits (up close and far away). If it does not, the answer is wrong.

3.2 Gauss' Law

Law 2 (Gauss' law). The flux of $\vec{E}$ through any closed surface is set by the charge inside it:

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0}. \quad (12)$$

Where Q_{enc}/ϵ_0 comes from. Start with a single point charge q at the centre of a sphere of radius r . On the sphere $\vec{E}$ is radial and $d\vec{a} = \hat{r} da$, so by Coulomb's law

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{q}{4\pi\epsilon_0 r^2} \oint_S da = \frac{q}{4\pi\epsilon_0 r^2} \cdot 4\pi r^2 = \frac{q}{\epsilon_0}.$$

The r^2 cancels, so every sphere around q gets the same flux. Three observations extend this:

- Any closed surface around q , of any shape: the same field lines that pierce the sphere must pierce it too, so its flux is still q/ϵ_0 .
- A charge outside the surface: each of its field lines enters once and leaves once, so it contributes zero net flux.
- Many charges: by superposition (Principle 1) the fluxes add, and only the enclosed charges contribute, giving Q_{enc}/ϵ_0 .

□

Proposition 5 (Differential form). *Apply the divergence theorem (Theorem 3) to the left side and write $Q_{\text{enc}} = \iiint_V \rho \, d\tau$:*

$$\iiint_V (\vec{\nabla} \cdot \vec{E}) \, d\tau = \frac{1}{\epsilon_0} \iiint_V \rho \, d\tau.$$

Since this holds for every volume V , the integrands match:

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}. \quad (13)$$

Definition 22 (Gaussian Surface). *The Gaussian surface just means an imaginary surface (nothing is physically there).*

Remark 13. *If there is no charge inside the Gaussian surface, the net flux is zero: every field line that enters on one side leaves on another.*

3.2.1 Spherical symmetry

Example (Point charge). *Surround a point charge q with a sphere S of radius r centred on it. Two facts reduce the integral to a product:*

- *By symmetry, $\vec{E} = E(r) \hat{r}$: the field is radial and has the same magnitude everywhere on S .*
- *By geometry, $d\vec{a} = \hat{n} \, da = \hat{r} \, da$ on a sphere.*

So $\vec{E} \cdot d\vec{a} = E \, da$, and

$$\oiint_S \vec{E} \cdot d\vec{a} = E \underbrace{\oiint_S da}_{4\pi r^2} = \frac{q}{\epsilon_0} \implies E = \frac{q}{4\pi\epsilon_0 r^2}.$$

3.2.2 Cylindrical symmetry

Example (Infinite line of charge). *This recovers the $z \ll L$ limit of the **line of charge example** in two lines. For a line with charge λ per unit length, take a coaxial⁵ cylinder of radius z and length d . On each end cap (a flat disk), the arrows of $\vec{E}$ lie in the disk, pointing outward from its centre like the spokes of a wheel: perpendicular to the line of charge and parallel to the cap. So $\vec{E} \cdot d\vec{a} = 0$ on the caps and only the curved side counts:*

$$E(2\pi z d) = \frac{\lambda d}{\epsilon_0},$$

so

$$E = \frac{\lambda}{2\pi\epsilon_0 z}. \quad (14)$$

The field falls off as $1/z$, not $1/z^2$.

Example (Finite versus infinite line). *Near its midpoint, a finite line has a weaker field than an infinite one with the same λ : the cylinder encloses the same charge, but some of the flux now leaks through the caps.*

⁵Sharing a common central axis.

3.2.3 Planar symmetry

Example (Infinite plane of charge). *For a plane with surface charge density σ , use a Gaussian pillbox: a short flat cylinder (like a hockey puck) that pierces the plane, half above and half below. It has two flat caps of area A , one on each side of the plane and parallel to it, joined by a side wall. By symmetry $\vec{E}$ points straight away from the plane on both sides, so it runs along the side wall (no flux there) and straight through each cap (flux EA each):*

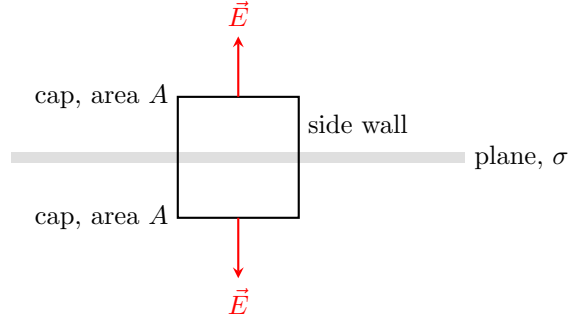


Figure 3: Side view of the pillbox straddling the plane.

$$2EA = \frac{\sigma A}{\epsilon_0},$$

so

$$E = \frac{\sigma}{2\epsilon_0}. \quad (15)$$

The field does not depend on the distance from the plane.

Example (Infinite plane via Coulomb's law). *The plane is still infinite, not a disk. We just describe it in polar coordinates (r', ϕ') centred directly below the field point at height z ; letting r' run from 0 to ∞ covers the whole plane. Sideways components cancel in pairs, so each piece $dq = \sigma r' dr' d\phi'$ contributes only its $\hat{z}$ part, via $\cos \psi = z/\sqrt{r'^2 + z^2}$ with ψ the angle between $\hat{\mathbf{r}}$ and $\hat{\mathbf{z}}$:*

$$\vec{E} = \hat{z} \int_0^{2\pi} d\phi' \int_0^\infty \frac{\sigma}{4\pi\epsilon_0} \frac{r'z}{(r'^2 + z^2)^{3/2}} dr' = \frac{\sigma z}{2\epsilon_0} \left[-\frac{1}{\sqrt{r'^2 + z^2}} \right]_0^\infty \hat{z} = \frac{\sigma}{2\epsilon_0} \hat{z}.$$

Distance drops out because up close each piece pushes hard but mostly sideways, while far away each piece is weak but pushes more directly upward. The two effects cancel exactly.

3.2.4 Why the fall-off depends on the geometry

The fall-off is how fast E decreases with the distance r from the source. The examples above give

$$\text{point charge: } E \propto \frac{1}{r^2}, \quad \text{infinite line: } E \propto \frac{1}{r}, \quad \text{infinite plane: } E \propto r^0 \text{ (constant).}$$

The exponent counts how much charge a cone of field lines can “see” as it grows. Picture a narrow cone opening from the observation point toward the source: its cross section at distance r grows like r^2 , which alone gives the $1/r^2$ dilution of a point charge. For extended sources, the charge inside the cone grows too:

- Line of charge: the cone intercepts a length $\propto r$, so $E \propto r/r^2 = 1/r$.
- Sheet of charge: the cone intercepts an area $\propto r^2$, so $E \propto r^2/r^2$, a constant. The infinite sheet has a uniform field.

Remark 14. *For two equal positive charges, no field lines cross the midplane: the field there is parallel to the plane and vanishes at the midpoint. Gauss’ law still holds, but without symmetry it cannot solve for $\vec{E}$.*

3.3 The Curl of $\vec{E}$

Example (Line integral of a point charge). *Nothing moves here. A point charge q sits fixed at the origin, and we compute the line integral of its field, $\int_{\vec{a}}^{\vec{b}} \vec{E} \cdot d\vec{l}$, along an arbitrary path from $\vec{a}$ to $\vec{b}$. (Times a test charge Q , this is the work the field would do on Q carried along that path.) In spherical coordinates $d\vec{l} = dr \hat{r} + r d\theta \hat{\theta} + r \sin \theta d\phi \hat{\phi}$, and since $\vec{E}$ is radial only the dr part survives:*

$$\int_{\vec{a}}^{\vec{b}} \vec{E} \cdot d\vec{l} = \frac{q}{4\pi\epsilon_0} \int_{r_a}^{r_b} \frac{dr}{r^2} = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_a} - \frac{1}{r_b} \right).$$

This depends only on the endpoints, so around a closed loop ($r_a = r_b$) it vanishes.

Theorem 7 (Electrostatic fields are curl-free). *For any electrostatic field,*

$$\oint \vec{E} \cdot d\vec{l} = 0, \quad \vec{\nabla} \times \vec{E} = 0. \quad (16)$$

Proof. Any static $\vec{E}$ is a sum of point-charge fields (superposition), so its loop integral is the sum of their loop integrals, each of which is zero. Stokes’ theorem then gives $\iint_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{a} = 0$ for every surface, so $\vec{\nabla} \times \vec{E} = 0$. $\square$

Remark 15. *Physically, $Q \oint \vec{E} \cdot d\vec{l}$ is the work done on a test charge around a closed loop. If it were nonzero, a charge could gain energy just by circling, violating energy conservation.*

3.4 Electric Potential

Definition 23 (Electric potential). *Since $\vec{\nabla} \times \vec{E} = 0$, the irrotational fields theorem guarantees a scalar V with*

$$\vec{E} = -\vec{\nabla}V. \quad (17)$$

The minus sign is convention: positive charges are pushed toward lower potential energy.

Remark 16. $\oint \vec{E} \cdot d\vec{l} = 0$ holds only in electrostatics. A changing magnetic field creates an electric field that circulates in loops (Faraday's law, $\vec{\nabla} \times \vec{E} = -\partial \vec{B} / \partial t$), so a charge can gain net energy going around. This is how a generator drives current around a closed circuit. Energy is still conserved: it comes from whatever is changing $\vec{B}$.

Proposition 6 (Potential from the field).

$$V(\vec{r}) = - \int_{\text{ref}}^{\vec{r}} \vec{E} \cdot d\vec{l}. \quad (18)$$

The path does not matter since the integral is path-independent⁶. By convention the reference is at infinity, unless the charge distribution extends to infinity.

Example (Point charge). Along a radial path,

$$V(r) = - \frac{q}{4\pi\epsilon_0} \int_{\infty}^r \frac{dr'}{r'^2} = \frac{q}{4\pi\epsilon_0 r}.$$

Proposition 7 (Potential of a distribution). By superposition,

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}')}{r} d\tau', \quad (19)$$

where $\rho(\vec{r}')d\tau' = dq$. Unlike the integral for $\vec{E}$, there is no $\hat{r}$, so computing V first and then $\vec{E} = -\vec{\nabla}V$ is often easier.

Law 3 (Poisson's equation). Combining $\vec{\nabla} \cdot \vec{E} = \rho/\epsilon_0$ with $\vec{E} = -\vec{\nabla}V$,

$$\nabla^2 V = -\frac{\rho}{\epsilon_0}. \quad (20)$$

3.5 Boundary Conditions

A surface charge σ makes $\vec{E}$ jump as you cross it. Gauss' law fixes the jump in the perpendicular part, and $\oint \vec{E} \cdot d\vec{l} = 0$ fixes the parallel part.

⁶Changing the reference point adds a constant to V , which leaves $\vec{E} = -\vec{\nabla}V$ unchanged (gauge freedom). Statement (ii) of the irrotational fields theorem.

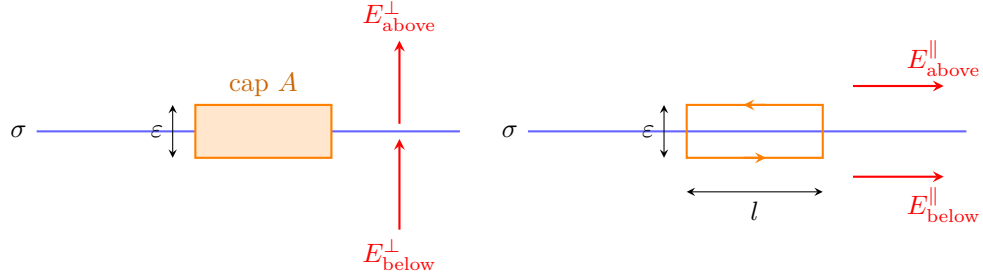


Figure 4: Side views of a charged surface. Left: a pillbox of cap area A and thickness ε gives the normal condition. Right: a thin rectangular loop of length l gives the tangential condition.

Proposition 8 (Normal component). *Take a pillbox of cap area A straddling the surface and shrink its thickness $\varepsilon \rightarrow 0$, so the side wall carries no flux. Only the caps count, and the charge inside is σA :*

$$E_{above}^\perp A - E_{below}^\perp A = \frac{\sigma A}{\epsilon_0} \implies E_{above}^\perp - E_{below}^\perp = \frac{\sigma}{\epsilon_0}.$$

Proposition 9 (Tangential component). *Take a thin rectangular loop of length l straddling the surface, again with $\varepsilon \rightarrow 0$ so the short sides contribute nothing. Since $\oint \vec{E} \cdot d\vec{l} = 0$,*

$$E_{above}^\parallel l - E_{below}^\parallel l = 0 \implies \vec{E}_{above}^\parallel = \vec{E}_{below}^\parallel.$$

Theorem 8 (Boundary condition on $\vec{E}$). *With $\hat{n}$ the unit normal pointing from below to above, the two results combine into*

$$\vec{E}_{above} - \vec{E}_{below} = \frac{\sigma}{\epsilon_0} \hat{n}. \quad (21)$$

The normal component jumps by σ/ϵ_0 ; the tangential component is continuous.

Remark 17. *Check with the infinite plane: $\vec{E} = +\frac{\sigma}{2\epsilon_0} \hat{n}$ above and $-\frac{\sigma}{2\epsilon_0} \hat{n}$ below, a jump of exactly σ/ϵ_0 .*

Theorem 9 (Boundary conditions on V). *The potential is continuous across the surface,*

$$V_{above} = V_{below}, \quad (22)$$

because $V_{above} - V_{below} = -\int_a^b \vec{E} \cdot d\vec{l}$ along a path straight through the surface, and as the path shrinks to zero length the integral vanishes ($\vec{E}$ stays finite). Its slope is not continuous: substituting $\vec{E} = -\vec{\nabla}V$ into Theorem 8 gives $\vec{\nabla}V_{above} - \vec{\nabla}V_{below} = -\frac{\sigma}{\epsilon_0} \hat{n}$, or with the normal derivative $\frac{\partial V}{\partial n} \equiv \vec{\nabla}V \cdot \hat{n}$,

$$\frac{\partial V_{above}}{\partial n} - \frac{\partial V_{below}}{\partial n} = -\frac{\sigma}{\epsilon_0}. \quad (23)$$

Remark 18. So V has a kink at a charged surface, like $|x|$ at $x = 0$: continuous, but its slope changes abruptly. $\vec{E}$, being the slope, jumps.

3.6 Work and Energy in Electrostatics

Proposition 10 (Potential of a test charge). *Bringing Q in from infinity against the field takes*

$$W = -Q \int_{\infty}^{\vec{r}} \vec{E} \cdot d\vec{l} = QV(\vec{r}). \quad (24)$$

So the electric potential V is the potential energy per unit charge, in $\text{J/C} = \text{V}$.

Proposition 11 (Energy of N point charges). *The first charge costs no work, since no field exists yet, and each later one feels the potential $V(\vec{r}_i)$ of those already placed, giving one term per pair:*

$$U = \frac{1}{2} \sum_{i \neq j} \frac{q_i q_j}{4\pi\epsilon_0 |\vec{r}_i - \vec{r}_j|} = \frac{1}{2} \sum_i q_i V(\vec{r}_i), \quad (25)$$

where the $\frac{1}{2}$ corrects for counting each pair twice.

Remark 19. U does not obey superposition, because it is quadratic in the charges rather than linear: doubling every charge doubles $\vec{E}$ but quadruples U . If two configurations with energies U_1 and U_2 are combined, the pair sum picks up new pairs (one charge from each), so $U = U_1 + U_2 + U_{\text{int}}$. In field language (Theorem 10), $E^2 = E_1^2 + E_2^2 + 2\vec{E}_1 \cdot \vec{E}_2$, and the cross term is the interaction energy.

Theorem 10 (Energy stored in the field). *For a continuous distribution,*

$$U = \frac{1}{2} \int \rho V d\tau = \frac{\epsilon_0}{2} \int_{\text{all space}} E^2 d\tau. \quad (26)$$

Definition 24 (Energy Density). *From Theorem 10, the energy per unit volume stored in the field is*

$$u = \frac{\epsilon_0}{2} E^2. \quad (27)$$

Proof. Using the energy of N point charges, $U = \frac{1}{2} \sum_i q_i V(\vec{r}_i)$. For a continuous distribution this becomes $U = \frac{1}{2} \int \rho V d\tau$. Substituting $\rho = -\epsilon_0 \nabla^2 V$ from Poisson's equation,

$$U = -\frac{\epsilon_0}{2} \int V \nabla^2 V d\tau.$$

To rewrite $V \nabla^2 V$, use the product rule $\partial_i(fg) = (\partial_i f)g + f(\partial_i g)$ with $f = V$ and $g = \partial_i V$, summed over i :

$$\underbrace{\sum_i \partial_i(V \partial_i V)}_{\vec{\nabla} \cdot (V \vec{\nabla} V)} = \underbrace{\sum_i (\partial_i V)^2}_{|\vec{\nabla} V|^2} + \underbrace{V \sum_i \partial_i \partial_i V}_{V \nabla^2 V} \implies V \nabla^2 V = \vec{\nabla} \cdot (V \vec{\nabla} V) - |\vec{\nabla} V|^2.$$

Substituting, the overall minus sign flips both terms:

$$U = -\frac{\epsilon_0}{2} \int [\vec{\nabla} \cdot (V \vec{\nabla} V) - |\vec{\nabla} V|^2] d\tau = \frac{\epsilon_0}{2} \int |\vec{\nabla} V|^2 d\tau - \frac{\epsilon_0}{2} \int \vec{\nabla} \cdot (V \vec{\nabla} V) d\tau.$$

The second integral vanishes. To see this, integrate over a huge sphere of radius R instead of all space, and let $R \rightarrow \infty$ at the end. The divergence theorem turns the integral into a flux through the sphere's surface, $\oint V \vec{\nabla} V \cdot d\vec{a}$. If the charge sits in a finite region, then far away $V \sim 1/R$ and $|\vec{\nabla} V| \sim 1/R^2$, so the integrand is $\sim 1/R^3$, while the area is $4\pi R^2$. The flux is therefore $\sim 1/R$, which goes to zero. With $\vec{E} = -\vec{\nabla} V$, $U = \frac{\epsilon_0}{2} \int E^2 d\tau$. $\square$

Remark 20 (Self-energy). *The two formulas do not always agree. The pair sum only counts the work of bringing different charges together ($i \neq j$); it ignores the work needed to build each point charge itself. The field integral counts everything, including that self-energy. So $\frac{\epsilon_0}{2} \int E^2 d\tau$ is always positive (since $E^2 \geq 0$), while the pair sum can be negative, e.g. $+q$ and $-q$ at distance d give $U = -\frac{q^2}{4\pi\epsilon_0 d}$. For a true point charge the self-energy is infinite: near it $E^2 \sim 1/r^4$, and $\int \frac{1}{r^4} r^2 dr$ blows up at $r = 0$. So use the pair sum for point charges and the field integral for continuous distributions.*

3.7 Conductors

Definition 25 (Conductor). *A conductor is a material with charges that are free to move (in a metal, the electrons). Placed in an external field, the free charges drift: negative ones pile up on one side and positive ones on the other, until the field they create, $\vec{E}_{\text{int}}$, exactly cancels $\vec{E}_{\text{ext}}$ inside. This happens almost instantly, and the result is electrostatic equilibrium.*

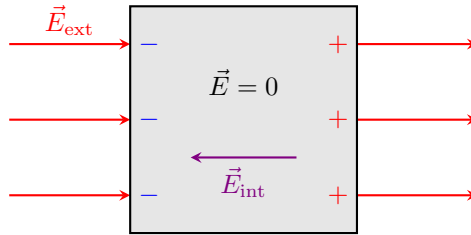


Figure 5: A conductor in an external field. The induced charges create $\vec{E}_{\text{int}}$, which cancels $\vec{E}_{\text{ext}}$ inside the metal. Outside, field lines end on the $-$ charges and start again from the $+$ charges.

Proposition 12 (Properties of a conductor in equilibrium). (i) $\vec{E} = 0$ inside, otherwise the free charges would still be moving.

(ii) $\rho = 0$ inside, since $\rho = \epsilon_0 \vec{\nabla} \cdot \vec{E} = 0$.

(iii) Any net charge therefore sits on the surface.

(iv) The conductor is an equipotential: for any two points in it, $V(\vec{b}) - V(\vec{a}) = -\int_{\vec{a}}^{\vec{b}} \vec{E} \cdot d\vec{l} = 0$.

(v) Just outside, $\vec{E}$ is perpendicular to the surface. A tangential part would push the surface charges along the surface, so they would not be in equilibrium.

Proposition 13 (Field at the surface of a conductor). *Theorem 8 with $\vec{E}_{\text{below}} = 0$ (inside the metal) gives, just outside,*

$$\vec{E} = \frac{\sigma}{\epsilon_0} \hat{n}, \quad (28)$$

or in terms of the potential, $\frac{\partial V}{\partial n} = -\frac{\sigma}{\epsilon_0}$. Where σ is large, the field outside is strong.

Remark 21. Compare the infinite sheet, $E = \sigma/2\epsilon_0$. A sheet sends half its flux up and half down; a conductor has no field inside, so all the flux goes out one side and the field is doubled.

Example (Neutral conductor near a charged rod). A neutral metal block sits near an insulating rod with uniform positive charge. Negative charge is induced on the side facing the rod and an equal positive charge on the far side. The negative charges are closer, and Coulomb's force falls off with distance, so the attraction beats the repulsion: the neutral conductor is pulled toward the rod.

Example (Conducting versus insulating disk). For a disk with uniform σ (an insulator), the disk is not an equipotential and the field lines are not perpendicular to it. That arrangement cannot last on a conductor: the field along the surface pushes charge outward, i.e. "downhill" in potential, and it piles up at the rim. So a charged conducting disk has a higher σ near its edge than near its centre, its surface is an equipotential, and its field lines leave it at right angles.

Remark 22. In general, charge on a conductor concentrates where the surface is most sharply curved (edges, corners, points). Since $E = \sigma/\epsilon_0$ there, the field is strongest at sharp points, which is how lightning rods work.

Remark 23 (Cavities). If a conductor has an empty cavity, the field inside the cavity is zero: its wall is an equipotential, and a solution of Laplace's equation with constant boundary values is constant (no maxima or minima, see the Laplace's equation section). This is a Faraday cage. If a charge q sits in the cavity, it induces $-q$ on the cavity wall and $+q$ on the outer surface (for a neutral conductor), and the outside cannot tell how the charge inside is arranged.

Example (Concentric conducting shells). A metal sphere of radius R with charge q sits inside a thick neutral metal shell of inner radius a and outer radius b .

- (i) A Gaussian sphere inside the shell's metal ($a < r < b$) has $\vec{E} = 0$ on it, so it encloses zero charge: the inner surface carries $-q$. The shell is neutral, so the outer surface carries $+q$. By symmetry each is uniform:

$$\sigma_R = \frac{q}{4\pi R^2}, \quad \sigma_a = -\frac{q}{4\pi a^2}, \quad \sigma_b = \frac{q}{4\pi b^2}.$$

- (ii) The field is $\frac{q}{4\pi\epsilon_0 r^2} \hat{r}$ for $r > b$ and $R < r < a$, and zero in both metals. Integrating in from infinity,

$$V(0) = - \int_{\infty}^b \frac{q dr}{4\pi\epsilon_0 r^2} - \int_a^R \frac{q dr}{4\pi\epsilon_0 r^2} = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{b} + \frac{1}{R} - \frac{1}{a} \right).$$

- (iii) Grounding the outer surface sets $V(b) = 0 = V(\infty)$, so there can be no field outside and $\sigma_b = 0$. The Gauss argument in (i) is unchanged, so σ_R and σ_a stay the same, and the first integral in (ii) disappears: $V(0) = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{R} - \frac{1}{a} \right)$.
- (iv) The outer surface now has zero charge; the $+q$ flowed away to the ground, leaving the shell with net charge $-q$. If the outer surface kept a charge Q' , the total charge inside a sphere just outside b would be Q' , giving $V(b) = \frac{Q'}{4\pi\epsilon_0 b} \neq 0$.

Example (Two cavities in a conductor). A neutral conducting sphere of radius R has two spherical cavities, of radii a and b , with point charges q_a and q_b at their centres.

- (i) Each cavity wall carries minus the charge inside it, spread uniformly since the charge is at the centre, and the outer surface carries the rest. The metal shields the outer surface from the cavities, so its charge spreads uniformly as on an isolated sphere:

$$\sigma_a = -\frac{q_a}{4\pi a^2}, \quad \sigma_b = -\frac{q_b}{4\pi b^2}, \quad \sigma_R = \frac{q_a + q_b}{4\pi R^2}.$$

- (ii) Outside, $\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q_a + q_b}{r^2} \hat{r}$, with r measured from the centre of the big sphere.
- (iii) Inside cavity a , $\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q_a}{r_a^2} \hat{r}_a$, with $\vec{r}_a$ measured from the centre of the cavity; the same holds for cavity b .
- (iv) The force on q_a and q_b is zero. Apart from its own field, q_a only feels the uniform shell of charge on its cavity wall, which produces no field inside it; everything beyond the wall is shielded by the metal.
- (v) Bringing q_c near the outside changes σ_R (it is no longer uniform) and the field outside, but nothing inside: σ_a , σ_b , the cavity fields and the forces on q_a and q_b stay the same. The conductor shields in both directions.

4 Potentials

4.1 Laplace's Equation

We want V because $\vec{E} = -\vec{\nabla}V$. The integral $V = \frac{1}{4\pi\epsilon_0} \int \frac{\rho}{r} d\tau'$ is exact but often impossible to do, and the multipole expansion is only an approximation. Often we do not even know ρ everywhere: we know the potentials of some conductors instead, and the charge on them is whatever it has to be. In the empty space between the charges, Poisson's equation simplifies.

Law 4 (Laplace's equation). *In a region with no charge,*

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0. \quad (29)$$

Remark 24. *Point charges, charged objects and conductors are where $\nabla^2 V = -\rho/\epsilon_0$ holds with $\rho \neq 0$. Everywhere else $\nabla^2 V = 0$, and the charges only enter as boundary conditions on that empty region.*

Remark 25 (The Laplacian compares a point with its surroundings). *In one dimension, $V'' = 0$ gives a straight line, $V = mx + b$, and then $V(x) = \frac{1}{2} [V(x+a) + V(x-a)]$: every point is the average of its neighbours. In general $\nabla^2 V$ measures how far a point sits from the average of its surroundings, and since $\nabla^2 V = -\rho/\epsilon_0$, that tells you the sign of the charge there.*

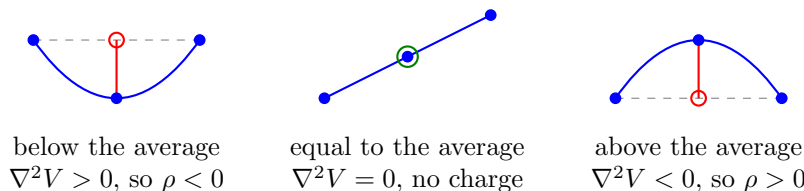


Figure 6: The middle point compared with the average of its two neighbours (open circle).

Theorem 11 (Mean value property). *If $\nabla^2 V = 0$, the value of V at any point equals its average over any sphere of radius R centred on that point (containing no charge):*

$$V(\vec{r}_0) = \frac{1}{4\pi R^2} \oint_{\text{sphere}} V da. \quad (30)$$

Corollary 11.1 (No local maxima or minima). *A solution of Laplace's equation has no local maxima or minima inside the region; extreme values occur only on the boundary. A maximum would be larger than every point around it, so it could not equal their average.*

Theorem 12 (Earnshaw's theorem). *A charged particle cannot be held in stable equilibrium by electrostatic forces alone.*

Proof. Stable equilibrium for a charge q needs its potential energy qV to have a local minimum at that point. But the other charges' V obeys Laplace's equation there, so it has no local minimum. Concretely, if $\frac{\partial^2 V}{\partial x^2} > 0$ (stable along x), then $\frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} < 0$, so the charge escapes sideways. For example, a charge between two equal charges is stable along the line joining them but slips out perpendicular to it. $\square$

Theorem 13 (Uniqueness theorem). *The solution of Laplace's equation in a volume is uniquely determined if V is specified on the whole boundary of that volume. Similarly, if the region contains conductors, $\vec{E}$ is uniquely determined once the charge density ρ and the total charge on each conductor are given.*

Remark 26. *The significance: any function that satisfies Laplace's equation and the boundary conditions is the answer, no matter how it was found. Guessing is allowed, which is what the method of images exploits.*

Proposition 14 (Method of relaxation). *On a grid, the mean value property says each cell equals the average of its neighbours. In two dimensions,*

$$V_{i,j} = \frac{1}{4} (V_{i+1,j} + V_{i-1,j} + V_{i,j+1} + V_{i,j-1}).$$

This gives a numerical method:

1. Initialize the grid with a guess for V .
2. Sweep through the grid, replacing each value by the average of its nearest neighbours, keeping the boundary values fixed.
3. Repeat until the values stop changing.

4.2 The Method of Images

Remark 27 (A solution in search of a problem). *Maxwell pointed out that the inverse problem is easier: given a potential, finding conductors that fit it is simpler than finding the potential for given conductors. Any equipotential surface of a known potential can be replaced by a conductor held at that potential, and by uniqueness the field outside it does not change.*

Example (Point charge above a grounded plane). *A charge q sits at height d above an infinite grounded conducting plane at $z = 0$. For $z > 0$ we need $\nabla^2 V = 0$ (apart from at q), $V = 0$ on the plane, and $V \rightarrow 0$ far away. Forget the plane and put an image charge $-q$ at $z = -d$ instead. Every point of the plane is equidistant from the two charges, so this pair gives $V = 0$ there, and it has the same charge in the region $z > 0$. By uniqueness it is the answer for $z \geq 0$:*

$$V(x, y, z) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{\sqrt{x^2 + y^2 + (z - d)^2}} - \frac{q}{\sqrt{x^2 + y^2 + (z + d)^2}} \right].$$

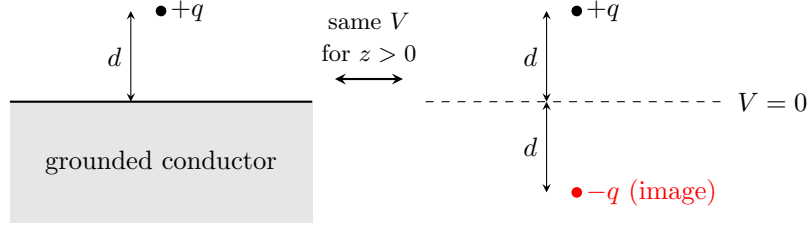


Figure 7: The real problem (left) and the image problem (right) have the same charge and the same boundary values in $z > 0$, so they have the same potential there.

Remark 28. *The image charge is not real; it is bookkeeping. Physically, the field comes from the negative surface charge induced on the plane, and the image reproduces its effect in the region we care about. The formula says nothing about $z < 0$, where $V = 0$ inside the conductor.*

Example (Two charges above a grounded plane). *Charges $+Q$ at height d and $+3Q$ at height $2d$ sit above a grounded plane. Each charge on its own is handled by one image, so by superposition we need two: $-Q$ at $-d$ and $-3Q$ at $-2d$. The method works whenever the problem is linear (superposition) and the boundary values come out right (uniqueness).*

4.3 Multipole Expansion

Example (Two charges on the z axis). *Place q_1 at $z = d/2$ and q_2 at $z = -d/2$, and measure θ from the z axis.*

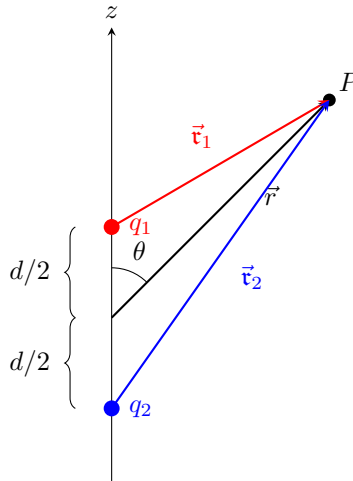


Figure 8: Two charges on the z axis and a far field point P at distance r and angle θ .

Then

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{\mathfrak{r}_1} + \frac{q_2}{\mathfrak{r}_2} \right).$$

To get $\mathfrak{r}_1$, write $\vec{\mathfrak{r}}_1 = \vec{r} - \frac{d}{2}\hat{z}$ and square it, using $\vec{r} \cdot \hat{z} = r \cos \theta$ (this is just the law of cosines):

$$\mathfrak{r}_1^2 = r^2 + \left(\frac{d}{2}\right)^2 - 2r\frac{d}{2}\cos\theta = r^2 + \left(\frac{d}{2}\right)^2 - rd\cos\theta.$$

For q_2 the source sits at $-\frac{d}{2}\hat{z}$, so the cross term flips sign. Both cases in one line, top sign for q_1 and bottom sign for q_2 :

$$\mathfrak{r}_{1,2}^2 = r^2 + \left(\frac{d}{2}\right)^2 \mp rd\cos\theta.$$

Writing $1/\mathfrak{r}_{1,2} = \frac{1}{r}(1 + \epsilon_{\mp})^{-1/2}$ with $\epsilon_{\mp} = \frac{d}{2r} \left(\frac{d}{2r} \mp 2\cos\theta \right)$ and expanding for $d \ll r$, $(1 + \epsilon)^{-1/2} = 1 - \frac{1}{2}\epsilon + \frac{3}{8}\epsilon^2 - \dots$, gives

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[\underbrace{\frac{q_1 + q_2}{r}}_{\text{monopole}} + \underbrace{\frac{(q_1 - q_2)d\cos\theta}{2r^2}}_{\text{dipole}} + \underbrace{\frac{(q_1 + q_2)d^2}{4} \frac{3\cos^2\theta - 1}{2r^3}}_{\text{quadrupole}} + \dots \right].$$

Each term gains a power of $1/r$, so $E \sim 1/r^2, 1/r^3, 1/r^4, \dots$

Remark 29. Setting $q_1 = -q_2$ kills the monopole term, leaving $V \sim 1/r^2$ and $E \sim 1/r^3$ at leading order.

Definition 26 (Dipole Moment). For charges $+q$ and $-q$ separated by $\vec{d}$ (pointing from $-q$ to $+q$), the dipole moment is

$$\vec{p} = q\vec{d}, \quad [\vec{p}] = \text{C} \cdot \text{m}. \quad (31)$$

For any distribution,

$$\vec{p} = \sum_i q_i \vec{r}_i = \int \vec{r}' \rho(\vec{r}') d\tau', \quad (32)$$

which reduces to $q\vec{d}$ for the pair. It measures how far apart the positive and negative charge sit, and sets the dipole term of the potential,

$$V_{\text{dip}} = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{r}}{r^2}. \quad (33)$$

Definition 27 (Ideal dipole). Take $q_1 = q$ and $q_2 = -q$. The monopole and quadrupole terms vanish, and the dipole term is $\frac{1}{4\pi\epsilon_0} \frac{qd\cos\theta}{r^2}$. Every term in the expansion carries a factor qd^n , which we can write as $(qd)d^{n-1}$. Now shrink $d \rightarrow 0$ while raising $q \rightarrow \infty$ so that the product qd stays fixed:

- The dipole term ($n = 1$) contains only qd , so it is unchanged.
- Every later term carries at least one extra factor of d , so it goes to zero.

(q must grow, otherwise the dipole term would vanish too.) What is left is a point object with dipole moment $\vec{p} = q\vec{d}$ and exactly $V = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{r}}{r^2}$.

Remark 30. The names count charges in powers of 2: monopole (1), dipole (2), quadrupole (4), octupole (8), and so on. Each is built from two opposite copies of the previous one, placed close together so the previous order cancels. For example, two opposite dipoles give a quadrupole with $V \sim 1/r^3$.

Theorem 14 (Multipole expansion). For r large compared with the size of the distribution,

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{Q}{r} + \frac{\vec{p} \cdot \hat{r}}{r^2} + \frac{\sum_{i,j} \hat{r}_i Q_{ij} \hat{r}_j}{r^3} + \dots \right), \quad (34)$$

with

$$Q = \int \rho(\vec{r}') d\tau', \quad \vec{p} = \int \vec{r}' \rho(\vec{r}') d\tau', \quad Q_{ij} = \int \frac{1}{2} (3r'_i r'_j - r'^2 \delta_{ij}) \rho(\vec{r}') d\tau'.$$

Griffiths writes the same series with Legendre polynomials; see Appendix A.1.

Remark 31. Each term is smaller than the previous one by a factor of roughly d/r , where d is the size of the charge distribution. Far away ($r \gg d$) the first nonzero term dominates and the rest are small corrections. Close up ($r \sim d$) the terms no longer shrink and the expansion is useless.

Remark 32 (Origin dependence). Shifting the origin by $\vec{a}$ changes $\vec{p} \rightarrow \vec{p} - Q\vec{a}$, so $\vec{p}$ is well defined only when $Q = 0$ (a neutral molecule, not an ion). In general only the first non-vanishing moment is origin-independent.

5 Electric Fields in Matter

6 Magnetostatics

6.1 Current Density

Definition 28 (Current density). The current I through a wire is not a local quantity, since a wire has finite thickness. The local version is the current density

$$\vec{J} = \frac{d\vec{I}}{da_\perp},$$

where $da_\perp$ is the differential cross section perpendicular to the flow.

Remark 33. Applying the divergence theorem (Theorem 3) to $\vec{J}$ is how conservation of charge gets written down: the charge inside V can only decrease by flowing out through S ,

$$\frac{d}{dt} \iiint_V \rho dV = - \oint_S \vec{J} \cdot d\vec{A} = - \iiint_V \vec{\nabla} \cdot \vec{J} dV \implies \frac{\partial \rho}{\partial t} = -\vec{\nabla} \cdot \vec{J}.$$

7 Magnetic Fields in Matter

8 Electrodynamics

A Appendix

Remark 34 (Compactified dimensions). *Suppose an extra dimension exists but is curled up, the way a flat sheet with coordinates (x, y) can be rolled into a cylinder whose second coordinate w is periodic with a small radius. At distances much smaller than that radius, field lines spread out into all the dimensions and the field falls off faster than $1/r^2$. At distances much larger than it, there is only room to spread out in the large dimensions and the usual $1/r^2$ is recovered. So a measured deviation from the inverse square law at short range would be evidence for extra dimensions.*

This has been tested for gravity by the Eöt-Wash torsion balance experiments (2004, 2007, 2020), which checked the $1/r^2$ law down to separations of $52\text{ }\mu\text{m}$. Newtonian gravity fit the data, ruling out gravitational strength Yukawa interactions with ranges above $38.6\text{ }\mu\text{m}$ at 95% confidence.

A.1 Multipole Expansion with Legendre Polynomials

Griffiths' form of the multipole expansion, with α the angle between $\vec{r}$ and $\vec{r}'$ and P_n the Legendre polynomials:

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \int (r')^n P_n(\cos \alpha) \rho(\vec{r}') d\tau'.$$

$P_0 = 1$ gives the monopole, $P_1 = \cos \alpha$ the dipole (since $r' \cos \alpha = \vec{r}' \cdot \hat{r}$), and $P_2 = \frac{1}{2}(3 \cos^2 \alpha - 1)$ the quadrupole.

B Useful Links

- Laplace relaxation demo:
<https://claude.ai/artifact/J2wPx8V3ajec8vDLsARnKX>
- Electrostatics simulation:
<https://icphysweb.z13.web.core.windows.net/simulation.html>